

REED PUTNAM PRACTICE · FALL 2026

Probability, Expectation & Games

Week 8 · expectation without distributions, games without trees

Thu Nov 19 / Fri Nov 20

The toolbox (1)

Linearity

$\mathbb{E}[X_1 + \dots + X_m] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_m]$ for **any** random variables, dependent or not. Use it through **indicators**: if X counts how many of $A_1, \dots, A_m$ occur, then $X = \sum \mathbf{1}_{A_i}$ and $\mathbb{E}[X] = \sum_i \mathbb{P}(A_i)$. **Choosing what to count** is the whole art.

- **Existence from averages**: some outcome has $X \geq \mathbb{E}[X]$, some has $X \leq \mathbb{E}[X]$. Integer $X \geq 0$ with $\mathbb{E}[X] < 1 \Rightarrow \mathbb{P}(X = 0) > 0$: pick at random, show **expected defects** < 1 .
- **Symmetry**: independent continuous variables are a.s. distinct and every order is equally likely: $\mathbb{P}(X_1 < \dots < X_n) = 1/n!$, $\mathbb{P}(X_i = \max_{j \leq i} X_j) = 1/i$.

The toolbox (2)

Games

A finite two-player perfect-information game with no draws has **exactly one** player with a winning strategy (Zermelo). Find it by **pairing/symmetry** (mirror every move), **strategy stealing** (a harmless extra move), or an **invariant** set $\mathcal{P}$: whoever moves to $\mathcal{P}$ wins.

- > **Conditioning:** $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$, $\mathbb{P}(A) = \sum_j \mathbb{P}(A \mid B_j) \mathbb{P}(B_j)$. If $\mathbb{E}[a_{k+1} \mid a_k]$ is affine in a_k , then $\mathbb{E}[a_k]$ obeys a **linear recursion**.
- > **Tail formulas:** $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) dt$; integer $X \geq 0$: $\sum_{k \geq 1} \mathbb{P}(X \geq k)$. So $\mathbb{E}[\min(U_1, \dots, U_i)] = 1/(i+1)$.

Example 1

Indicators and existence

Every graph with m edges has a bipartite subgraph with at least $m/2$ edges.

→ work it on the board (~3 min)

Example 2

Conditioning and induction: Putnam 2002 B1

Shanille hits her first free throw, misses her second, and thereafter hits each shot with probability equal to the fraction of shots hit so far. Find the probability she hits exactly 50 of her first 100.

→ work it on the board (~3 min)

Example 3

Strategy stealing: Chomp

An $m \times n$ chocolate bar ($mn > 1$) has a poisoned square at the bottom left. Players alternately pick a square and eat it together with every square above it, to its right, or both. Whoever eats the poison loses. Show the first player has a winning strategy.

→ work it on the board (~3 min)

How to recognize it

- > “Expected value” of a count: **indicators + linearity**; one $\mathbb{P}(A_i)$ by symmetry, then add.
- > A max or a stopping time: rewrite as a sum, or **condition on the first step** and recurse.
- > “Some object exists” with a numerical bound: random construction, $\mathbb{E}[\text{defects}] < 1$.
- > Games: a **symmetry** to mirror? a harmless extra move (**steal**)? an **invariant** to restore?

Pitfalls

- > Linearity needs no independence; products, variances and $\mathbb{P}(A \cap B)$ do: justify them.
- > $\mathbb{E}[1/X] \neq 1/\mathbb{E}[X]$, $\mathbb{E}[\max] \neq \max \mathbb{E}$; “ties have probability 0” must be said.
- > Pairing fails if the mirror move is illegal or just played; stealing says *who*, not *how*.

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.